Constantinos Paschalis

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Portfolio

Profile

MSc Energy Science and Technology student at ETH Zürich specializing in energy systems modeling and optimization. Experience in techno-economic analysis, heat pump systems, and CO_2 transport modeling across academic and industry settings.

Education

ETH Zürich, Zurich — 2024 – Present
MSc Energy Science and Technology

Texas A&M University, Qatar — 2024
Energy and Sustainability Summer School

University of Cyprus, Nicosia — 2020 – 2024
BSc Mechanical Engineering — 9.06/10 (Top Graduate)

Experience

EnergyIntel Services Ltd, Nicosia — Jun 2025 – Aug 2025
R&D Intern — Modeled thermal energy storage systems and evaluated fluid selection using multi-criteria analysis.

Texas A&M University, USA — Jun 2023 – Jul 2023
Research Assistant (HVAC) — Performed hardware-in-the-loop and simulation-based analysis of HVAC systems under varying climate conditions.

University of Cyprus, Nicosia — May 2022 – Sep 2022
Research Assistant — Developed computational models and performed simulation-based analysis.

Projects

CO_2 Transport Network Resilience – ETH Zürich
Built techno-economic model to evaluate resilience under failure scenarios; quantified cost–reliability trade-offs.

Mega Heat Pump Economic Optimization – ETH Zürich
Developed thermo-economic optimization model integrating thermodynamic cycle modeling and electricity market pricing; analyzed system profitability.

Household Energy Optimization – ETH Zürich
Developed optimization models integrating electricity prices and CO_2 constraints to minimize cost.

Skills

Programming: Python, MATLAB
Modeling: ANSYS, SolidWorks
Core: Energy Systems, Thermodynamics, Optimization
Languages: Greek (Native), English (Fluent)

Awards

ETEK, AHK, Elysse Irrigation LTD (2024) — Top Mechanical Engineering Graduate (9.06/10)
MUSKITA Awards (2022, 2023) — Highest academic performance (2nd & 3rd year)

Interests

Energy systems modeling, optimization, energy markets, sustainable technologies